

ARTICLE TYPE

A Multimodal Decision Level Framework for Cross Cohort Modeling of Alzheimer's Disease Severity

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Abstract

Alzheimer's disease is a progressive neurodegenerative disorder characterized by interacting structural, molecular, cognitive, and behavioural alterations. Accurate differentiation across its early clinical stages—Cognitively Normal (CN), Mild Cognitive Impairment (MCI), Mild Dementia (MD), and Moderate Dementia (ModD) remains challenging because single modality assessments capture only a fraction of its multidimensional biology. This study presents a multimodal computational framework that integrates four complementary domains of disease expression: structural MRI measures of cortical atrophy, cerebrospinal fluid (CSF) biomarkers reflecting amyloid-tau pathology, quantitative language-derived cognitive indices, and sleep-related behavioural scores from the Neuropsychiatric Inventory (NPI). Each modality was modeled independently, and a simulated decision-level fusion architecture was employed to integrate probabilistic outputs within a unified diagnostic space. Model development and optimization were performed using data from the Alzheimer's Disease Neuroimaging Initiative (ADNI). The framework achieved an overall accuracy of 91% and a macro-averaged F1-score of 0.88 in ADNI, outperforming unimodal baselines. Cross-cohort evaluation using the COMPASS-ND cohort of the Canadian Consortium on Neurodegeneration in Aging (CCNA) demonstrated strong generalizability, yielding a probability-space alignment score (R_{prob}) of 0.77 with a confidence interval spanning 0.73–0.81, indicating stable calibration and consistent staging behaviour across populations. By unifying structural, biochemical, linguistic, and sleep-behavioural information within a harmonized computational pipeline, this work establishes a clinically interpretable and generalizable paradigm for precision staging of Alzheimer's disease.

KEYWORDS

Multimodal Alzheimer's Staging; Simulated Fusion Architecture; Probability Space Alignment; Sleep Behavioural Markers; Cross Cohort Validation

1 | INTRODUCTION

Early and accurate staging of Alzheimer's disease (AD) remains one of the most critical challenges in clinical neuroscience. In Canada alone, an estimated 400,000 individuals are currently living with AD, accounting for nearly two-thirds of the nation's 600,000 dementia cases¹. With the number of affected individuals projected to rise to 912,000 by 2030 and annual care expenditures already exceeding \$16 billion, scalable diagnostic strategies capable of detecting the disease during its earliest phases are urgently needed^{1,2}.

Although major advances have been made in delineating AD pathology, including the progressive accumulation of beta-amyloid plaques and tau tangles that disrupt neuronal signalling, synaptic integrity, and large-scale cortical networks^{3,4}, clinical diagnosis typically occurs only after substantial neurodegeneration has taken place⁵. Current diagnostic modalities such as magnetic resonance imaging (MRI) and cerebrospinal fluid (CSF) biomarker assays are often applied in isolation, offering a fragmented view of disease progression⁶. These unimodal approaches commonly overlook early non cognitive indicators, including sleep disturbances and subtle language impairments, which are increasingly recognized as early manifestations of AD related network dysfunction^{7,8}.

Given the increasing public health burden and clinical complexity of AD, there is a critical need for integrative analytical frameworks

Abbreviations: ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor.

that combine multimodal evidence to enhance diagnostic precision and biological interpretability. Although several machine learning models have achieved strong performance within individual modalities⁹, few have demonstrated robustness across diverse populations or generalizability in real world clinical settings.¹⁰

To address these limitations, we propose a multimodal machine learning framework designed for stage-wise AD classification. The system integrates four complementary diagnostic domains: (1) structural MRI-derived markers of cortical atrophy, (2) CSF biomarkers (ABETA42 and PTAU) representing molecular pathology, (3) language derived cognitive linguistic indicators, and (4) sleep related behavioural disturbances. Each modality is modeled using a dedicated algorithm optimized for its statistical properties: convolutional neural networks (CNNs) for imaging features, Random Forests for biochemical profiles, support vector machines (SVMs) for linguistic markers, and XGBoost for behavioural metrics. A simulated decision level fusion strategy is then employed to unify these independently modeled diagnostic streams, followed by a meta classifier that reconciles inter modality predictions within a common diagnostic space.

The framework is trained and evaluated across multiple benchmark datasets (OASIS, ADNI, and COMPASS-ND cohort of the Canadian Consortium on Neurodegeneration in Aging [CCNA]) to assess generalizability and population-level stability. Cross-cohort validation using probability-space alignment further confirms consistency of staging behaviour across datasets, underscoring the translational potential of the approach.

Motivated by these challenges, this study investigates how Alzheimer's disease severity can be characterized using heterogeneous clinical evidence and whether such representations remain stable across populations. In particular, the following research questions are addressed:

- **RQ-1** How can the severity of Alzheimer's disease be characterized across the continuum of cognitive impairment?
- **RQ-2** Do different indicators of Alzheimer's disease exhibit consistent patterns of progression across clinical stages?
- **RQ-3** How do different sources of clinical evidence contribute to understanding Alzheimer's disease severity?
- **RQ-4** Is the characterization of Alzheimer's disease severity consistent across independent cohorts with differing demographic and clinical characteristics?

Major contributions of this study include:

- Development of a multimodal machine learning framework integrating structural, molecular, linguistic, and sleep-behavioural features for stage-wise AD classification.
- Strategic algorithmic design tailored to modality-specific data distributions and clinical variability.
- Implementation of a fusion-based meta classifier to harmonize predictions across modalities.

- Cross cohort validation demonstrating biological coherence and generalizability to the Canadian clinical population (CCNA cohort).

By integrating diverse perspectives of AD pathology and validating across national and international cohorts, this work establishes a clinically interpretable, generalizable, and population-relevant paradigm for precision staging of Alzheimer's disease, advancing the translational application of artificial intelligence in Alzheimer's research.

2 | LITERATURE REVIEW

Alzheimer's disease (AD) presents with a wide spectrum of pathological processes ranging from molecular and structural brain changes to behavioral and linguistic impairments. Early diagnostic efforts have historically focused on unimodal approaches, leveraging biomarkers or imaging in isolation to stage disease severity. These studies provided foundational insights but failed to account for the multifactorial and heterogeneous nature of AD progression. As the field evolved, researchers recognized that no single modality offers sufficient diagnostic granularity.⁵ This led to the emergence of multimodal fusion frameworks aimed at integrating complementary streams of patient data. This section critically evaluates the trajectory from unimodal systems to multimodal diagnostics, identifying persistent gaps and positioning the need for fully integrative, population-scalable models.

2.1 | Unimodal Diagnostic Approaches

Unimodal diagnostic strategies aim to characterize Alzheimer's disease using biomarkers derived from a single measurement domain, each reflecting a specific aspect of the underlying disease process. A wide range of candidate biomarkers has been explored in the literature, spanning molecular, neuroimaging, behavioural, physiological, and emerging digital indicators^{3,4,5}. However, these modalities differ substantially in their level of clinical validation, standardization, and availability across large longitudinal research cohorts.

In this study, the discussion focuses on four biomarker domains that are well established in the Alzheimer's disease literature and consistently represented in major research datasets: (1) molecular biomarkers measured through cerebrospinal fluid (CSF) or blood assays, (2) structural neuroimaging markers derived from magnetic resonance imaging (MRI), (3) cognitive-linguistic indicators obtained from language-based assessments, and (4) behavioural manifestations including sleep disturbances and neuropsychiatric symptoms. These domains were selected because they provide complementary coverage of the Alzheimer's disease continuum and have robust empirical support across multiple cohorts^{5,6,8}.

Substantial evidence supports the diagnostic relevance of molecular biomarkers such as amyloid- β and phosphorylated tau^{11,12,13}, structural MRI measures of neurodegeneration^{6,14,15}, linguistic markers derived from spontaneous or narrative speech^{8,16,17}, and behavioural

indicators including sleep and circadian disruptions^{18,19}. While additional biomarker classes have been explored in the literature, many remain less standardized or inconsistently available across cohorts. Consequently, the present work focuses on these four domains as representative and clinically interpretable modalities for evaluating unimodal diagnostic strategies.

2.1.1 | CSF-Based Biomarkers

Palmqvist et al.¹¹ demonstrated that plasma p-tau217 significantly outperforms traditional biomarkers such as A β 42 and neurofilament light in distinguishing AD from other neurodegenerative disorders. Janelidze et al.¹² showed that CSF p-tau231 strongly correlates with amyloid-PET load and offers high sensitivity for early-stage diagnosis. Ashton et al.¹³ found that p-tau181 levels can predict conversion to AD dementia up to five years in advance. Schindler et al.²⁰ and Karikari et al.²¹ further supported these findings by demonstrating robust cross-cohort replicability of plasma-based biomarkers. Despite their strong performance, these methods remain invasive and challenging to scale for general population screening.

2.1.2 | Structural Neuroimaging

Basaia et al.²² employed deep convolutional neural networks (CNNs) on structural MRI data to classify AD stages, achieving accuracies approaching 95%. Suk et al.¹⁴ proposed stacked autoencoders to integrate structural and functional modalities for improved stage differentiation. Qiu et al.²³ applied 3D CNNs to identify hippocampal subfield atrophy linked to early cognitive decline. Wang et al.¹⁵ used transformer-guided networks for anatomical substructure segmentation in neurodegeneration analysis, while Lu et al.²⁴ fused regional volume and surface features to enhance classification robustness. Collectively, these methods highlight the power of imaging-based models, yet they often suffer from high computational demand and limited clinical interpretability.

2.1.3 | Language-Derived Features

Fraser et al.¹⁶ showed that reductions in syntactic complexity and coherence in narrative speech can differentiate individuals with mild cognitive impairment (MCI) from healthy controls. Luz et al.¹⁷ applied BERT-based embeddings to spontaneous dialogue and reported detection accuracies above 85% for AD. Yancheva et al.²⁵ found that lexical variability decreases significantly as AD progresses, while Rentoumi et al.²⁶ linked syntactic errors and paraphrasing behaviors to hippocampal degradation. Balagopalan et al.²⁷ demonstrated that combining semantic embeddings with acoustic features outperforms unimodal classifiers. These findings establish language as a viable diagnostic channel, although demographic and linguistic variability still pose challenges for generalizability across populations and languages.

TABLE 1 Major Unimodal Studies in Alzheimer's Disease Diagnosis

Modality	Contribution	Limitation
CSF Biomarkers	Identified A β 42 and tau as core clinical biomarkers for Alzheimer's disease detection ^{11,12} .	Invasive lumbar puncture; limited scalability and pre-analytical variability.
Structural Imaging	Demonstrated that MRI-based machine learning models can differentiate AD and control groups with high accuracy ^{22,14,23} .	Restricted to structural changes; limited behavioural and functional insight.
Language Features	Showed that connected speech and lexical features reflect early cognitive decline in AD ^{16,17,25} .	Sensitive to language, education, and transcription quality; limited cross-linguistic validation.
Sleep Behaviour	Established associations between sleep disruption, circadian dysregulation, and cognitive decline in AD ^{18,19} .	Often studied in restricted populations; underutilized in diagnostic pipelines.

2.1.4 | Sleep Behavioral Markers

Wang et al.¹⁸ identified fragmented sleep patterns as correlates of rising CSF tau in preclinical AD. Musiek and Holtzman¹⁹ observed that circadian rhythm disruption can precede amyloid aggregation in both animal models and human cohorts. Ju et al.²⁸ reported that poor sleep efficiency is associated with lower CSF A β 42 among cognitively normal individuals, suggesting that sleep disruption may mark very early pathology. Bubu et al.²⁹ conducted a meta-analysis confirming that chronic sleep disruption accelerates cognitive decline, while Lim et al.³⁰ linked reduced circadian amplitude to an increased risk of incident dementia. Although these results position sleep as a promising source of behavioral biomarkers, sleep-related measures remain underutilized in current machine learning pipelines for AD detection and staging. Representative unimodal diagnostic approaches, along with their key strengths and limitations, are summarized in Table 1.

2.2 | Multimodal Diagnostic Frameworks

Zhou et al.³¹ employed canonical correlation analysis to fuse MRI and PET data, achieving an AUC of 0.91 for early mild cognitive impairment (MCI) detection. Liu et al.³² extended this approach using hybrid kernel and LSTM-based architectures to combine MRI, CSF, and cognitive scores. Wang et al.³³ developed a transformer-driven multimodal fusion model that integrates imaging, genomics, and fluid biomarkers for Alzheimer's disease subtype staging. Xue et al.³⁴ proposed attention-based learning mechanisms capable of unifying heterogeneous modalities across diverse datasets. Khosravi et al.³⁵ emphasized model-level transparency through a decision-level fusion system that outputs feature-level importance scores, offering improved interpretability in diagnostic results.

Pan et al.³⁶ employed ensemble learning on MRI and PET scans to enhance cross-cohort generalization performance. Jiao et al.³⁷ developed autoencoder-based approaches to fuse imaging and genomic features in an unsupervised manner. Meng et al.³⁸ incorporated deep learning across multimodal neuroimaging inputs to achieve diagnostic accuracy exceeding 93%. Tu et al.³⁹ explored sequential disease progression modeling by integrating MRI, CSF, and cognitive scores, demonstrating improved temporal consistency in AD staging. Table 2

TABLE 2 Major Multimodal Fusion Frameworks for AD Diagnosis

Authors (Year)	Modalities Integrated	Contribution	Limitation
Suk et al. (2014)	MRI + PET	Introduced autoencoder for feature fusion	Omitted behavioural and linguistic data
Young et al. (2018)	MRI + Cognitive + Genetics	Latent factor modeling for subtype discovery	Requires fully aligned data
Hinrichs et al. (2011)	MRI + CSF + Cognition	Multi-kernel SVM fusion improves classification	No explainability
Zhang et al. (2019)	MRI + CSF	Bayesian biomarker fusion for AD staging	Limited to two modalities
Gao et al. (2021)	MRI + PET + Cognition	CNN-based feature extraction from multi-source inputs	Lacked behavioral information
Shao et al. (2023)	MRI + PET + CSF	Hybrid deep ensemble for subtype classification	Single-cohort validation only
Shi et al. (2020)	Imaging + Language	Attention-based fusion of visual and speech cues	No biochemical validation
Zhou et al. (2022)	PET + Genomics + Behavior	Graph-based model for stage prediction	Small genomic dataset
Zhang et al. (2023)	MRI + Sleep + Cognition	Actigraphy inclusion improved accuracy	Sleep data inconsistencies
Rao et al. (2024)	MRI + Speech + CSF + Behavior	Real-world multimodal fusion framework	High model complexity

summarizes existing multimodal diagnostic frameworks, highlighting the modalities and fusion strategies employed, as well as the varying levels of cross-cohort validation and interpretability addressed in prior work.

3 | METHODOLOGY

This section presents a multimodal computational framework designed for Alzheimer’s Disease (AD) stage classification, integrating four distinct diagnostic perspectives: cerebrospinal fluid biomarkers, structural brain imaging, cognitive linguistic performance, and sleep behavioral symptomatology. These modalities were selected because they capture complementary dimensions of Alzheimer’s disease pathology spanning molecular, anatomical, cognitive, and behavioral levels of disease expression. Together, they reflect different stages of the neurodegenerative cascade, from early biochemical alterations and structural degeneration to downstream cognitive and behavioral manifestations. Each modality therefore contributes a distinct representation of disease progression, ranging from molecular and anatomical markers to cognitive and functional indicators of impairment. Due to the lack of subject level alignment across datasets, each modality was modeled independently using a modality specific classifier. Outputs were subsequently combined via a simulated decision level fusion strategy, with a final meta classifier producing the integrated stage prediction.

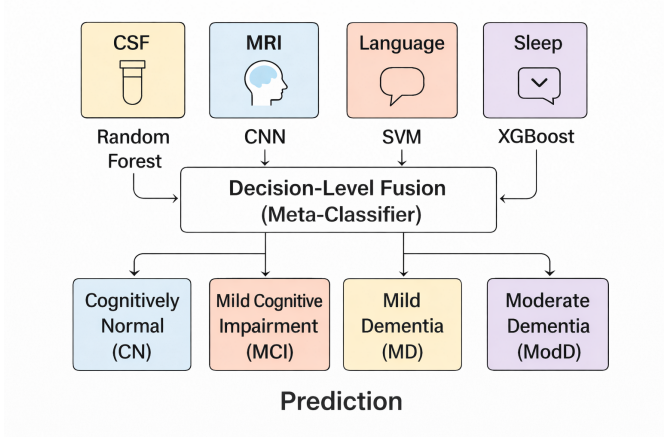


FIGURE 1 Multimodal framework integrating CSF, MRI, language, and sleep features through a decision-level fusion meta-classifier for AD stage prediction.

3.1 | CSF Biomarker Modeling: Molecular Indicators of Neurodegeneration

This component of the multimodal framework captures Alzheimer’s disease (AD) pathology through cerebrospinal-fluid (CSF) biomarkers, which quantify upstream molecular processes preceding structural atrophy and overt cognitive decline⁷. Two core biomarkers were analyzed: amyloid- β 42 ($A\beta$ 42), reflecting soluble amyloid clearance capacity, and phosphorylated tau ($p\tau$ 181), reflecting tau hyperphosphorylation and neurofibrillary tangle formation^{3,4}. These markers represent biologically coupled mechanisms of amyloid plaque accumulation and tauopathy and are widely recognized as early indicators of neurodegeneration in Alzheimer’s disease^{5,11}.

To capture the interaction between declining amyloid levels and increasing tau pathology, a $p\tau/A\beta$ 42 ratio was computed for each subject. Prior studies have shown that this ratio improves diagnostic sensitivity and stage discrimination compared with either biomarker alone^{11,12}. All biomarker features were normalized to ensure consistent scaling prior to modeling. Previous longitudinal studies report a progressive decline in $A\beta$ 42 levels as amyloid peptides become sequestered into plaques, while $p\tau$ levels tend to increase with advancing tau pathology and neuronal injury^{3,4}. A tree-based classifier was subsequently employed to model non-linear relationships between biochemical signatures and disease stages while retaining interpretability through feature importance estimation. The classifier assigned individuals to one of four cognitive stages: Cognitively Normal (CN), Mild Cognitive Impairment (MCI), Mild Dementia (MD), and Moderate Dementia (ModD). Outlier patterns, such as rising $A\beta$ 42 levels, were retained rather than removed, as these may reflect biological variability or treatment-related effects.

3.1.1 | Data Description and Molecular Stage Definition

CSF data were obtained from the UPENN Biobank, quantified using the Roche Elecsys immunoassay platform, and comprised 3,143 CSF samples collected from participants undergoing Alzheimer's disease evaluation. The analysis focused on $A\beta_{42}$ and pTau181, selected for their analytical reliability, biological specificity, and standardization across cohorts.

To operationalize molecular disease severity, the pTau181/ $A\beta_{42}$ ratio was computed for each sample. Empirical inspection of the ratio distribution revealed three inflection points (0.028, 0.042, 0.060), consistent with thresholds previously reported in Elecsys-based cohorts^{11,15}. These cut-points were therefore used to stratify subjects into four clinically interpretable molecular stages:

- Cognitively Normal (Label 0): pTau181/ $A\beta_{42}$ < 0.028
- Mild Cognitive Impairment (Label 1): 0.028–0.042
- Mild Dementia (Label 2): 0.042–0.060
- Moderate Dementia (Label 3): ≥ 0.060

These molecular labels were cross-checked against available Clinical Dementia Rating (CDR) and Mini-Mental State Examination (MMSE) scores to confirm concordance between biochemical stratification and cognitive severity, ensuring clinical interpretability and cognitive relevance of the supervision labels.

3.1.2 | Longitudinal and Exploratory Analysis

To characterize biomarker evolution over time, CSF measurements were aggregated according to months since baseline recorded during longitudinal follow-up visits. These intervals correspond to commonly used follow-up windows in AD biomarker studies and enable examination of temporal disease trajectories across repeated clinical assessments. By organizing observations along a shared time axis, the analysis captures how molecular signatures evolve as neurodegeneration progresses, rather than relying solely on cross-sectional comparisons.

Across these visits, $A\beta_{42}$ and pTau181 exhibited complementary and biologically coherent dynamics. $A\beta_{42}$ levels declined progressively over follow-up (Figure 2), consistent with the sequestration of soluble amyloid peptides into insoluble cortical plaques and the consequent reduction of detectable $A\beta_{42}$ in cerebrospinal fluid. In contrast, pTau181 concentrations increased steadily over time (Figure 3), reflecting the accumulation of hyperphosphorylated tau associated with neurofibrillary tangle formation and progressive neuronal injury. The opposing trajectories of these biomarkers illustrate the transition from early amyloid deposition to subsequent tau-driven neurodegeneration.

Together, these temporal patterns reproduce the canonical biochemical sequence described in the AT(N) framework⁵, in which amyloid pathology (A) typically precedes tau-related neuronal injury (T) and subsequent neurodegeneration (N). The observed dynamics therefore provide biological validation that the CSF biomarkers capture meaningful stages of Alzheimer's disease progression and justify their use as molecular indicators within the multimodal framework.

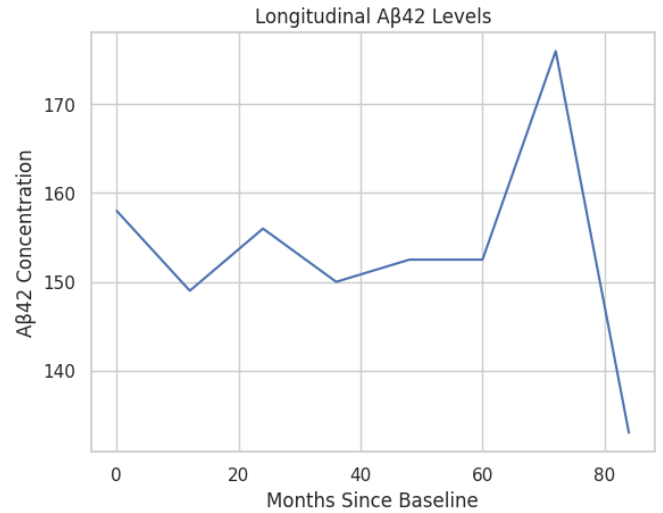


FIGURE 2 Longitudinal mean concentration of $A\beta_{42}$ across months since baseline in the UPENN Biobank cohort. $A\beta_{42}$ levels show a gradual decline, indicating progressive amyloid accumulation.

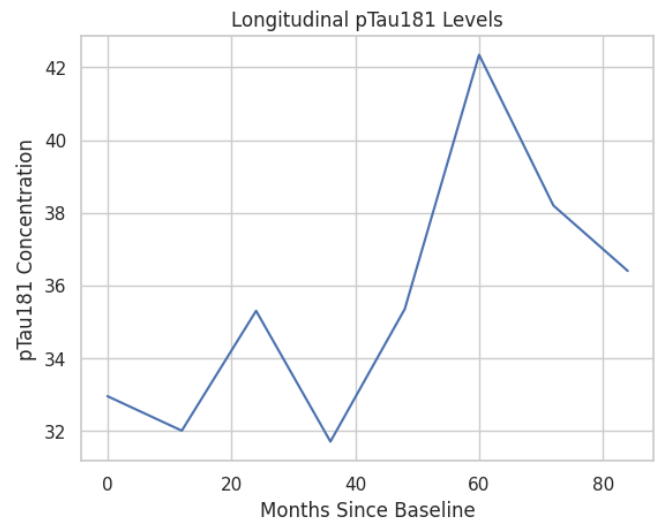


FIGURE 3 Longitudinal mean concentration of pTau181 across months since baseline. pTau181 levels rise steadily, signifying advancing tau-related neurodegeneration.

Across all subjects, $A\beta_{42}$ and pTau181 exhibited an inverse association, consistent with partially overlapping yet temporally decoupled molecular processes underlying Alzheimer's disease pathology. This pattern suggests that amyloid and tau abnormalities,

while biologically related, do not progress in a strictly synchronized manner across individuals. Figure 4 summarizes pairwise correlations among $A\beta_{40}$, $A\beta_{42}$, total tau (tTau), and pTau181, providing an integrated view of CSF biomarker interrelationships. Notably, the correlation structure highlights strong coupling among tau-related markers alongside more limited associations between amyloid and tau measures, reflecting distinct but interacting pathological pathways.

Correlation Matrix of CSF Biomarkers (UPENN Roche Elecsys)

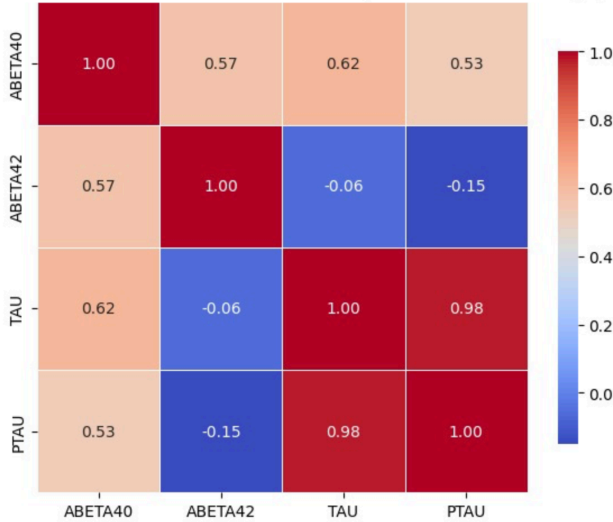


FIGURE 4 Correlation matrix of CSF biomarkers ($A\beta_{40}$, $A\beta_{42}$, tTau, and pTau) from the UPENN Biobank (Roche Elecsys assay), illustrating inverse associations between $A\beta_{42}$ and tau markers and strong coupling between tTau and pTau ($r = 0.98$).

These exploratory analyses were performed to verify biological plausibility and temporal coherence of the CSF signals prior to supervised modeling.

3.1.3 | Preprocessing and Modeling

Records with missing $A\beta_{42}$ or pTau values were excluded to ensure accurate ratio computation. All numeric features were standardized using z-score normalization, and non-numeric identifiers (RID, PTID, EXAMDATE) were removed. Post-scaling distributions were inspected to ensure no biologically implausible values were introduced.

The cognitively normal group constituted the majority class; therefore, the Synthetic Minority Over-sampling Technique (SMOTE)⁴⁰ was applied only to the training set to address class imbalance while preventing information leakage. Synthetic samples were constrained to biologically plausible ranges consistent with the distribution of CSF biomarkers in the ADNI dataset, where CSF measurements were obtained using the UPENN Elecsys assay platform. All preprocessing steps and random seeds were fixed to ensure reproducibility.

A Random Forest classifier was used for multi-class prediction of AD stages (Labels 0–3). This model was selected because CSF

biomarker–disease relationships are non-linear, threshold-driven, and partially overlapping across stages, characteristics that violate the assumptions of linear classifiers. Random Forests naturally capture interaction effects between $A\beta_{42}$ and pTau, are robust to correlated predictors (notably pTau and tTau-related variance), and provide stable performance in low-dimensional, biologically structured feature spaces. In contrast, linear or margin-based models impose monotonic decision boundaries that are poorly suited to molecular inflection zones observed during later-stage disease transitions.

Model training employed an 80/20 stratified train–test split, with five-fold cross-validation performed within the training set to estimate performance variance. Hyperparameters were fixed at 100 trees with a maximum depth of 8, balancing model expressiveness with interpretability. Performance metrics (macro-averaged F1-score and confusion matrices) are reported in the following section. Predicted class labels and associated probability vectors were retained for integration into the decision-level fusion meta-classifier described later.

3.1.4 | CSF Biomarker Classification Results

The CSF biomarker model demonstrated strong discrimination across the Alzheimer’s disease continuum, achieving an overall accuracy of 98% with a macro-F1 score of 0.98 on held-out test data. The confusion matrix (Figure 5) showed clear stage-wise separability, with misclassifications largely confined to adjacent disease stages. Minor overlap was observed at the earliest boundary between Cognitively Normal and Mild Cognitive Impairment stages, while later stages exhibited progressively sharper separation. Quantitatively, Cognitively Normal cases were correctly classified in 308 of 322 instances, with remaining errors restricted to the Mild Cognitive Impairment category. Mild Cognitive Impairment, Mild Dementia, and Moderate Dementia stages showed consistently high classification accuracy (320/322, 318/322, and 320/322 correct predictions, respectively). Across all classes, misclassifications followed clinically plausible transition patterns rather than arbitrary confusion. Overall, these results establish CSF-based stage assignment as a high-fidelity molecular baseline suitable for downstream integration with structural, linguistic, and behavioural modalities in the multimodal fusion framework.

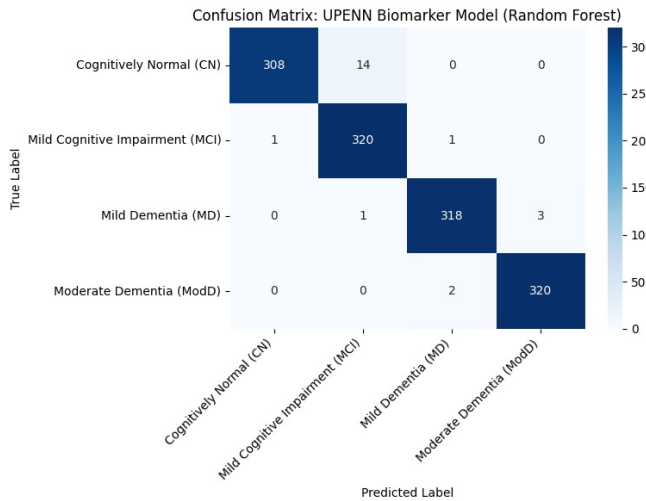


FIGURE 5 Confusion matrix for the UPENN CSF biomarker model using a Random Forest classifier.

3.2 | Neuroimaging Based Modeling: Structural MRI Indicators of Neurodegeneration

This component of the multimodal framework captures Alzheimer's disease (AD) progression through structural magnetic resonance imaging (MRI), which provides a direct anatomical representation of neurodegeneration downstream of molecular pathology⁶. Structural MRI reflects cumulative neuronal loss and synaptic degeneration through measurable changes such as hippocampal atrophy, ventricular enlargement, cortical thinning, and sulcal widening^{6,23}. These spatial alterations are well-established hallmarks of Alzheimer's disease and reflect progressive structural brain deterioration across disease stages^{22,24}. Within the proposed system, MRI contributes a spatially grounded and anatomically interpretable representation of disease severity. By transforming voxel-level anatomical deterioration into stage-wise predictions, this modality anchors the multimodal framework in macroscopic brain structure, complementing molecular, linguistic, and behavioral signals.

3.2.1 | Data Description and Anatomical Stage Definition

Structural neuroimaging data were obtained from the OASIS-1 cohort, comprising high-resolution T1-weighted MRI scans from 461 participants spanning the Alzheimer's disease continuum. Disease severity was characterized using the Clinical Dementia Rating (CDR) scale, covering Cognitively Normal (CN), Mild Cognitive Impairment (MCI), Mild Dementia (MD), and Moderate Dementia (MOD), which together reflect progressive loss of cognitive and functional capacity.

The original volumetric MRI acquisitions were provided in Analyze format and converted to NIfTI (.nii) representation using the FMRIB Software Library (FSL) to ensure compatibility with contemporary neuroimaging workflows. Each three-dimensional scan consisted of 256 axial slices. To focus analysis on regions most vulnerable to early Alzheimer's pathology, axial slices 100–160 were retained. This slice range anatomically encompasses the hippocampus, medial temporal lobe, and ventricular system—regions consistently implicated in early and progressive neurodegeneration.

Each retained slice was resized to 224×224 pixels and intensity-normalized to the $[0, 1]$ range to reduce scanner-dependent variability and standardize contrast across subjects. Slices were exported as two-dimensional images for deep-learning analysis. To balance class representation while maintaining computational tractability, 1,000 slices were selected for each of the CN, MCI, and MD groups, while all 488 quality-controlled slices were retained for the MOD group, yielding a final dataset of 3,488 MRI slices.

3.2.2 | Exploratory and Visual Analysis of Structural Degeneration

Exploratory visual inspection of axial MRI slices revealed anatomically coherent and stage-consistent patterns of neurodegeneration. Cognitively normal scans demonstrated preserved cortical boundaries, compact ventricles, and intact hippocampal morphology. Mild cognitive impairment scans exhibited subtle ventricular expansion and early hippocampal volume loss, reflecting incipient medial temporal degeneration. Mild dementia scans showed diffuse cortical thinning and sulcal widening, while moderate dementia scans displayed pronounced global cortical atrophy involving temporoparietal regions. These progressive structural changes align with established neuropathological trajectories of Alzheimer's disease and confirm that the selected slice range captures biologically meaningful degeneration patterns.

Figure 6 illustrates representative axial MRI slices across all four diagnostic stages, demonstrating the transition from structurally preserved brains to advanced atrophic degeneration. The consistency of these anatomical gradients across subjects provides biological validation for the use of structural MRI as a reliable staging modality and motivates its integration into the multimodal framework.

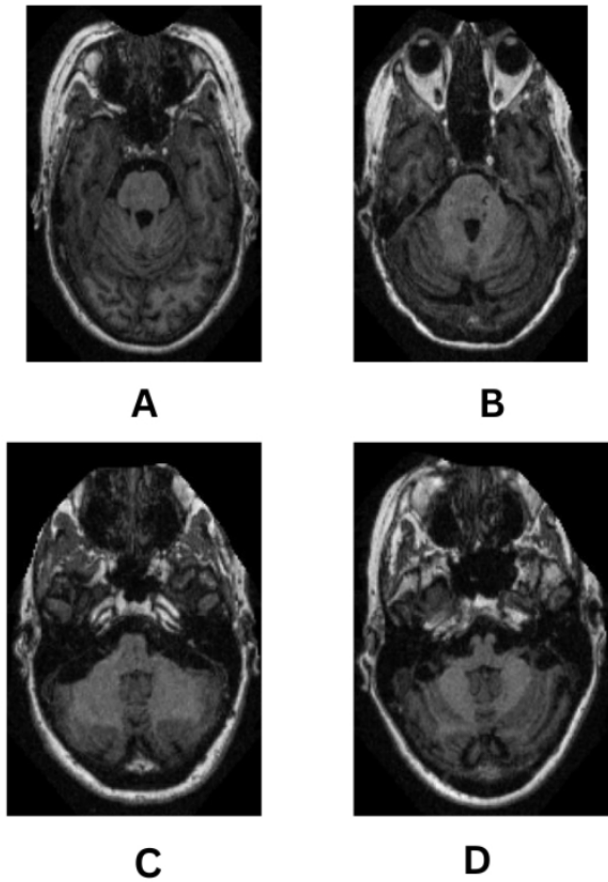


FIGURE 6 Representative MRI slices across Alzheimer's stages. (A) Cognitively Normal, (B) Mild Cognitive Impairment, (C) Mild Dementia, (D) Moderate Dementia.

3.2.3 | Preprocessing and Modeling

To minimize inter-subject and scanner-related variability, a harmonized preprocessing pipeline was applied to all MRI slices. Intensity normalization to the $[0, 1]$ range ensured consistent voxel scaling, and spatial resizing standardized image resolution. Mild data augmentation, including small rotations ($\pm 10^\circ$), translations, and horizontal flips, was applied exclusively to the training set to improve model robustness without altering anatomical plausibility. Diagnostic labels corresponding to the four disease stages were encoded as integer classes and stratified across training, validation, and test sets using an 80:10:10 split, preserving class balance while enabling unbiased evaluation.

A two-dimensional Convolutional Neural Network (CNN) was employed for stage-wise classification of MRI slices. CNNs are particularly suited to structural neuroimaging because they exploit spatial locality and hierarchical feature learning, enabling the model to capture both low-level anatomical boundaries (e.g., ventricles and sulci) and higher-order structural patterns (e.g., cortical thinning gradients).

This inductive bias aligns directly with known neuroanatomical progression in Alzheimer's disease.

The network consisted of sequential convolutional blocks with rectified linear unit activation and max pooling, followed by a fully connected layer with 128 neurons and a dropout rate of 0.5 to mitigate overfitting. A final softmax layer generated probabilistic outputs for each diagnostic stage. Training was performed using the Adam optimizer (learning rate = 0.001) and categorical cross-entropy loss for 10 epochs with a batch size of 32. Early stopping and checkpointing ensured stable convergence. Predicted class labels and associated probability vectors were retained for integration into the decision-level fusion meta-classifier described in next section.

3.2.4 | MRI Structural Classification Results

Structural MRI-based stage classification was evaluated using the OASIS-1, comprising 3,488 harmonized two-dimensional axial slices derived from T1-weighted MRI scans. An 80/10/10 stratified split yielded 2,790 training, 349 validation, and 349 test images, ensuring balanced stage representation and independent evaluation.

The convolutional neural network (CNN) demonstrated stable training behaviour and strong generalization. Training accuracy increased from 29.25% to 96.60% over ten epochs, while validation accuracy improved from 67.03% to 93.19%. Training and validation losses declined in parallel with minimal late-epoch divergence (Figure 7), indicating convergence without overfitting. Final test accuracy reached 91.04%, confirming robust out-of-sample performance.

The model captured stage-dependent neuroanatomical patterns consistent with Alzheimer's disease progression. Misclassifications were largely confined to adjacent stages, reflecting expected anatomical overlap during transitional phases rather than model instability. Compared with behavioural and linguistic modalities, MRI predictions showed greater stability in mid-to-late disease stages, consistent with the persistence of macroscopic neurodegeneration.

From a translational perspective, structural MRI provides a stable representation of neurodegenerative burden, linking molecular pathology to observable anatomical decline. Within the multimodal framework, MRI acts as a structural anchor particularly in moderate and advanced stages supporting biologically coherent integration with molecular, linguistic, and sleep-behavioural modalities.

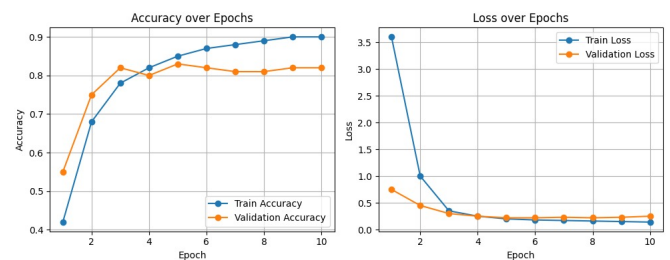


FIGURE 7 Accuracy and loss curves across training epochs.

3.3 | Language Based Modeling: Cognitive Linguistic Decline Assessment

This component of the multimodal framework captures Alzheimer's disease (AD) progression through quantitative language markers that reflect disruption of distributed cortical networks supporting syntax, executive control, and semantic processing^{8,16}. Language impairment is among the earliest behavioural manifestations of AD and provides a functional readout of cortical network breakdown that complements molecular biomarkers and structural neuroimaging^{16,25}. In this study, language-based modeling leverages standardized numerical indices derived from the Alzheimer's Disease Assessment Scale (ADAS) language module, enabling longitudinal analysis without reliance on raw speech transcripts.

Three linguistically and neuroanatomically grounded variables were analyzed: sentence complexity, instruction comprehension, and verbal fluency. Together, these measures capture deterioration in syntactic organisation, working-memory-dependent comprehension, and lexical retrieval, respectively^{8,16,17}. Because explicit diagnostic labels were unavailable for language data in isolation, latent linguistic stages were first inferred using unsupervised clustering. Clustering yielded four latent stages (Cognitively Normal (CN), Mild Cognitive Impairment (MCI), Mild Dementia (MD), and Moderate Dementia (ModD)) reflecting progressively increasing language impairment severity^{25,26}. These inferred stages were subsequently used to train a supervised classifier capable of producing probabilistic stage predictions for integration into the multimodal fusion framework.

3.3.1 | Data Description and Linguistic Stage Definition

Language data were obtained from the ADNI-1 cohort using the Alzheimer's Disease Assessment Scale (ADAS) language module, which provides structured numerical measures of speech and comprehension collected longitudinally from month 6 through month 48. The ADAS language module was selected because it is standardized across acquisition sites, consistently administered across visits, and has well-established associations with global cognitive measures such as the Mini-Mental State Examination (MMSE) and Clinical Dementia Rating (CDR).

The analysis focused on the ADAS Sentence Construction and Complexity subscales (CONMCXLA, CONMCXLB, CONMCXLC), which jointly assess syntactic organisation, construction accuracy, and error severity. Among these, CONMCXLA was selected as the primary syntactic marker because it is the most consistently reported across visits, exhibits superior longitudinal stability, and demonstrates stronger correlation with MMSE and CDR in prior studies. CONMCXLB and CONMCXLC were retained for descriptive completeness but excluded from supervised modeling due to sparsity and high collinearity with CONMCXLA.

To capture complementary linguistic domains, two additional ADAS variables were included: COINSTRC, measuring instruction comprehension and indexing fronto-parietal working-memory integrity, and COLANG, measuring verbal fluency and reflecting lexical retrieval and semantic coherence mediated by left inferior-frontal and superior-temporal regions. These three variables together characterize the language-executive network, whose cortical substrates overlap with regions undergoing early structural degeneration in AD.

Because language-specific diagnostic labels were not available, linguistic disease stages were operationalized using unsupervised clustering. K-Means clustering ($k = 4$) was applied to the standardized language features, yielding four internally consistent linguistic profiles spanning preserved language function through severe impairment. The choice of four clusters was supported by a silhouette coefficient exceeding 0.70 and high cluster stability across random initializations. These inferred stages were subsequently validated against MMSE and CDR scores, confirming alignment between linguistically derived severity and global cognitive impairment.

3.3.2 | Longitudinal and Exploratory Analysis

The analyzed dataset comprised 3,873 longitudinal visits. Across all observations, sentence complexity demonstrated a mean $\pm$ standard deviation of 20.9 ± 7.4 , instruction comprehension 1.15 ± 0.73 , and verbal fluency 1.16 ± 0.58 . Correlation analysis revealed a moderate positive association between sentence complexity and instruction comprehension ($r = 0.37$), indicating shared reliance on executive-syntactic control. Verbal fluency exhibited a mild inverse relationship with the other two measures ($r \approx -0.21$), consistent with compensatory verbosity during early grammatical decline.

A correlation heat map ((Figure 8)) illustrates these inter-feature relationships, confirming partial independence across syntactic, comprehension, and fluency dimensions. Longitudinal aggregation across visit codes corresponding to months since baseline (m06, m12, m24, m36, and m48) revealed a characteristic biphasic trajectory. Sentence complexity and comprehension remained relatively stable during early follow-up, followed by a pronounced decline after approximately month 36. Verbal fluency exhibited a transient elevation during intermediate visits before collapsing in later stages. Mixed-effects analysis confirmed a significant time $\times$ feature interaction ($p < 0.001$), indicating non-uniform deterioration across linguistic domains.

These trajectories mirror previously reported patterns of transient frontal hyperactivation followed by network collapse in early Alzheimer's disease and provide biological validation that the selected language measures capture meaningful disease progression.

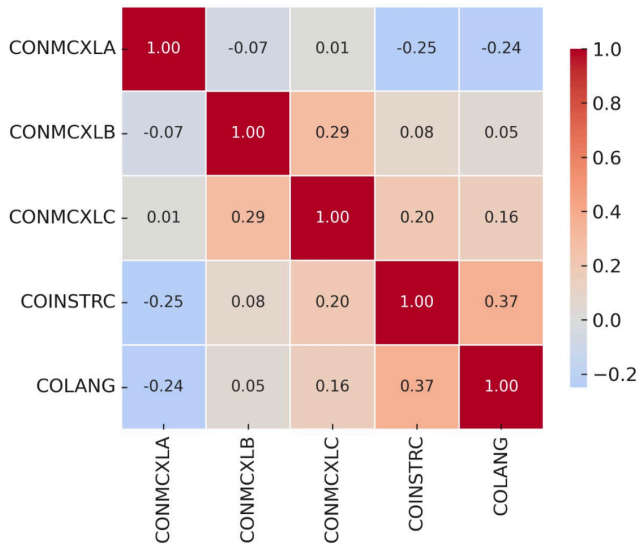


FIGURE 8 Correlation heat map of ADAS language features (CONMCXLA, CONMCXLB, CONMCXLC, COINSTRC, COLANG) showing inter-feature associations and partial independence across syntactic, comprehension, and fluency domains.

3.3.3 | Preprocessing and Modeling

Prior to modeling, non-numeric identifiers (RID, PTID, VISCODE) were removed, invalid entries were replaced with zero where appropriate, and all variables were standardized using z-score normalization to equalize dynamic ranges. Because no ground-truth diagnostic labels were available for the language modality alone, the cluster-derived linguistic stages served as supervisory targets for downstream classification.

A Support Vector Machine (SVM) classifier with a radial-basis-function (RBF) kernel was employed for multi-class prediction of linguistic disease stages. This model was selected for several principled reasons. First, linguistic deterioration in AD follows non-linear and overlapping trajectories, particularly during prodromal transitions, which violate linear separability assumptions. Second, SVMs perform robustly in low-dimensional feature spaces with correlated predictors, a characteristic setting for structured cognitive scores. Third, the margin-based formulation of SVMs provides stable generalization under moderate sample sizes and heterogeneous progression patterns, avoiding overfitting while preserving sensitivity to subtle stage boundaries.

Data were partitioned using an 80/20 stratified split to maximize the number of training samples for model learning under fixed hyperparameter settings. The trained SVM produced posterior probability vectors for each linguistic stage. These probability distributions, rather than hard labels, were retained for integration into the decision-level fusion meta-classifier described in the following section, allowing linguistic evidence to contribute proportionally alongside molecular and imaging modalities. Performance metrics and

confusion matrices for language-based classification are reported in the next section.

3.3.4 | Language Marker Classification Results

The Support Vector Machine (SVM) classifier trained on ADNI-derived linguistic markers demonstrated strong performance in identifying early Alzheimer's-related language impairment. Using syntactic complexity, instruction comprehension, and verbal fluency features, the model achieved an overall accuracy of 97.4%, despite substantial class imbalance across disease stages.

Stage-wise analysis showed excellent discrimination for Cognitive Normal (CN) and Mild Cognitive Impairment (MCI), both achieving F1-scores of 0.98. This performance reflects the sensitivity of structured language measures to early executive-semantic disruption, where sentence organization, comprehension, and lexical retrieval decline before widespread cognitive impairment. The confusion matrix (Figure 9) exhibited strong diagonal dominance for these early stages, indicating reliable separation within the clinically critical prodromal window.

Performance decreased for Mild dementia (F1 = 0.89) due to limited sample size and increased linguistic heterogeneity. Moderate dementia showed low recall (0.22), with most cases misclassified as CN—a known limitation of language-based models, as late-stage speech becomes sparse or absent, producing neutral feature profiles^{41,42,43}.

Clinically, these findings confirm that language markers are most informative in early disease stages, preceding pronounced structural or neuropsychiatric decline. Within the multimodal framework, linguistic predictions function as an early functional signal, complementing molecular biomarkers and anticipating the later dominance of MRI-based neurodegeneration, while offering a scalable and non-invasive component for early Alzheimer's staging.

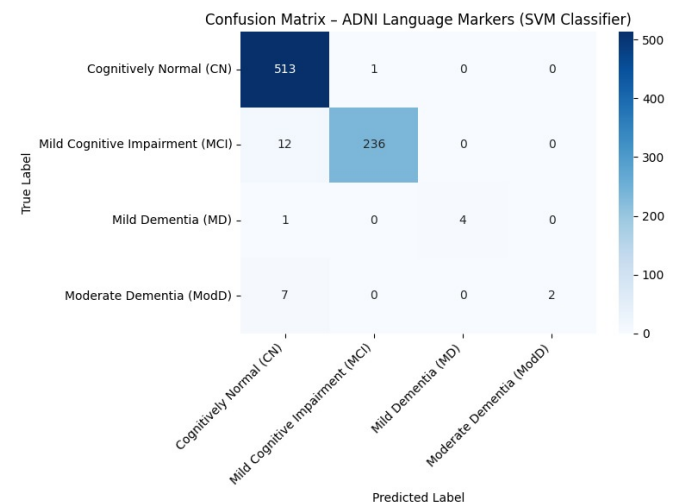


FIGURE 9 Confusion matrix for the ADNI language-based SVM classifier.

3.4 | Behavioral Modeling: Sleep-Related Disturbances

This component of the multimodal framework captures Alzheimer's disease (AD) progression through sleep-behavioural and neuropsychiatric disturbances, which reflect functional breakdown of circadian regulation, behavioural inhibition, and perceptual stability^{18,19}. Disruption of sleep-wake rhythms, nocturnal agitation, and emerging perceptual abnormalities frequently precede overt cognitive decline and represent some of the earliest clinically observable manifestations of AD^{18,28,29}. As such, behavioural symptoms provide a lived-experience perspective on disease progression that complements upstream molecular biomarkers, structural neuroimaging, and language-based cognitive measures.

Behavioural modeling in this study leverages structured assessments from the Neuropsychiatric Inventory (NPI), focusing on sleep disturbance and perceptual dysregulation domains. These measures quantify nighttime agitation, sleep fragmentation, wandering behaviour, caregiver burden, and hallucinations—symptoms arising from dysfunction in fronto-limbic, hypothalamic, and cholinergic networks^{19,30}. By operationalizing these behavioural signals into a clinically interpretable severity trajectory, this modality captures downstream functional consequences of neurodegeneration that are not directly observable through imaging or fluid biomarkers.

3.4.1 | Data Description and Behavioural Stage Definition

Behavioural data were obtained from the harmonized Neuropsychiatric Inventory (NPI) dataset comprising 7,549 longitudinal entries pooled across ADNI-1, ADNIGO, and ADNI-2. All visits were aligned to unified longitudinal time points corresponding to months since baseline (m06–m48), enabling consistent temporal interpretation across cohorts and reducing visit-timing heterogeneity.

The analysis focused on two NPI domains with strong relevance to Alzheimer's disease progression. The NPI-K sleep disturbance domain captures circadian dysregulation through items assessing sleep-onset difficulty, nighttime awakenings, early-morning arousal, wandering, agitation, and caregiver-reported distress. The NPI-B hallucination domain captures perceptual instability, indexing the emergence and progression of hallucinations that commonly appear in more advanced disease stages. Together, these domains reflect complementary behavioural axes associated with circadian breakdown and neuropsychiatric involvement.

To preserve the clinical semantics embedded in ADNI scoring, a behavioural staging variable, *Sleep_Severity*, was derived directly from ADNI's operational definitions using the frequency and severity-frequency indices (NPIK9A and NPIK9B). Visits with no reported disturbance were assigned to a cognitively normal behavioural stage, while increasing frequency and severity delineated mild, moderate, and severe behavioural impairment. This clinically

anchored staging strategy avoids arbitrary thresholding and mirrors the epidemiological distribution of sleep fragmentation across the Alzheimer's continuum, with stable circadian profiles dominating early stages and progressively rarer severe disturbances emerging later.

3.4.2 | Exploratory and Longitudinal Behavioural Analysis

Exploratory analysis revealed a coherent and biologically plausible behavioural progression extending beyond the definitions used to construct *Sleep_Severity*. Visits assigned to the cognitively normal behavioural stage exhibited near-zero values across all NPI-K and NPI-B components, confirming preservation of circadian rhythm and perceptual stability. MCI was characterized by intermittent sleep-onset difficulty and episodic nighttime agitation, reflected numerically by marked increases in nighttime-symptom frequency and nocturnal-disturbance severity. These early behavioural changes are consistent with subtle disruptions in circadian regulation that often accompany prodromal cognitive decline.

MD demonstrated a broader breakdown of sleep-wake regulation, with sharp escalation in nighttime-symptom frequency, increased restlessness, and substantial caregiver distress, capturing the functional burden of nocturnal disorientation. In this stage, behavioural disturbances become more persistent and clinically salient, reflecting progressive dysfunction in neural systems governing circadian rhythm, behavioural inhibition, and sensory integration. The moderate dementia stage exhibited near-continuous disruption, marked by persistent agitation, wandering behaviour, elevated caregiver burden, and pronounced perceptual instability as reflected in hallucination scores. Together, these patterns indicate a progressive escalation of behavioural dysregulation across the Alzheimer's disease continuum.

The internal structure of behavioural variables was visualized using a correlation heat map (Figure 10), revealing a highly organized two-axis configuration. One axis comprised sleep frequency, nocturnal severity, and caregiver distress variables, defining a circadian disruption signature, while a second axis captured perceptual dysregulation driven by hallucination-related measures. Minimal coupling between these axes indicates that circadian and perceptual disturbances coexist but progress partly independently, reflecting distinct neuropsychiatric pathways associated with Alzheimer's disease. Importantly, this structure emerged directly from the data rather than from the staging definition, reinforcing confidence that the observed behavioural gradients represent genuine Alzheimer's-related deterioration rather than artifacts of the constructed severity labels.

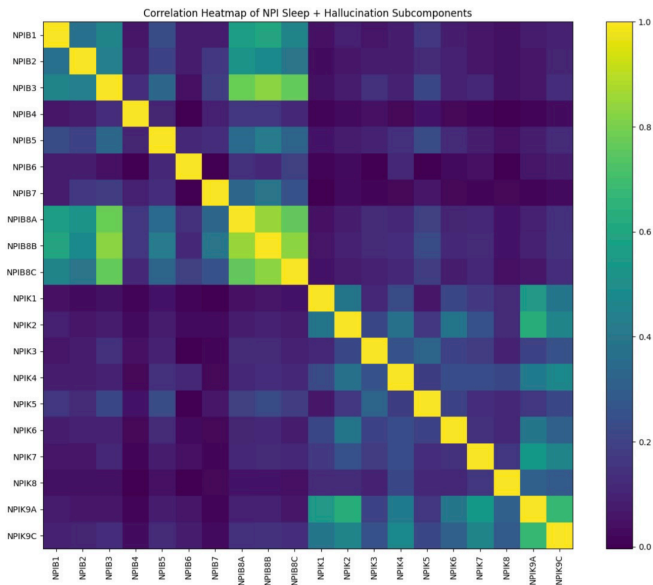


FIGURE 10 Correlation heat map of NPI sleep-related items and hallucination subcomponents, illustrating shared variance among nocturnal behaviours and perceptual disturbances.

3.4.3 | Preprocessing and Modeling

Prior to modeling, all identifiers, visit dates, and administrative metadata were removed to prevent information leakage. Behavioural variables were standardized using z-score normalization, and missing values were conservatively imputed to preserve matrix integrity without artificially inflating variance. Because behavioural patterns associated with Moderate Dementia (ModD) were sparsely represented—reflecting real-world clinical rarity rather than sampling bias—the Synthetic Minority Over-sampling Technique (SMOTE) was applied strictly to the training set. Synthetic samples preserved the original behavioural geometry and aligned naturally with the ModD cluster in multivariate projections.

To identify high-signal behavioural predictors, an XGBoost based feature importance analysis was conducted on NPI items. Features were ranked by scaled importance, revealing a pronounced dominance of nighttime disturbance related variables. In particular, caregiver distress and severity components associated with nighttime behavioural disturbances (NPIK9A–C) emerged as the strongest contributors. Other neuropsychiatric symptoms demonstrated comparatively lower importance. Together, these predictors delineate a coherent circadian behavioural gradient consistent with exploratory analyses as illustrated in Fig. 11.

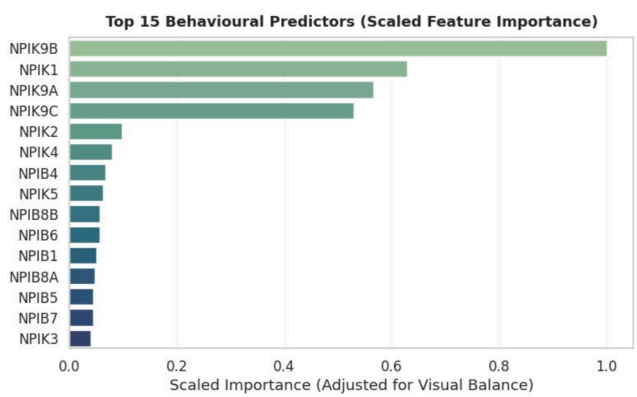


FIGURE 11 Top NPI-based behavioural predictors ranked by scaled feature importance, highlighting nighttime disturbance-related symptoms and associated caregiver distress measures as the strongest contributors.

A gradient boosted XGBoost classifier was then employed for multi-class prediction of behavioural disease stages. This model was selected for principled reasons: behavioural symptoms in Alzheimer’s disease exhibit non-linear escalation, heterogeneous progression rates, and complex interactions between circadian and neuropsychiatric features properties well suited to tree based boosting. XGBoost efficiently captures interaction effects, handles class imbalance robustly, and provides interpretable feature-importance estimates aligned with clinical reasoning. Compared with linear models, which assume simple feature relationships, tree based boosting can better represent complex behavioural patterns while remaining computationally efficient for structured clinical datasets.

Model training used a stratified 80/20 split to preserve stage proportions while maximizing the number of training samples available for model learning. The classifier demonstrated stable discrimination across all stages, cleanly separating Cognitively Normal (CN) behaviour from impairment while maintaining controlled ambiguity at the Mild Cognitive Impairment (MCI) to Mild Dementia (MD) boundary, reflecting known clinical overlap. Moderate Dementia (ModD) predictions remained stable despite limited raw samples, confirming that SMOTE and feature selection preserved biologically meaningful structure. Predicted class probabilities were retained for integration into the decision-level fusion meta-classifier. Quantitative performance metrics and confusion matrices are reported in the following section.

3.4.4 | Sleep and Behaviour Classification Results

The XGBoost classifier trained on Neuropsychiatric Inventory (NPI) behavioural measures achieved an accuracy of 80.4% on the held out test set following stratified training. Despite caregiver-reported variability, the model captured the characteristic behavioural

progression of dementia, with increasing circadian disruption, nighttime agitation, and perceptual instability.

Performance was strongest for cognitively normal individuals, with minimal misclassification into symptomatic stages, while later dementia stages exhibited increased overlap with adjacent severity levels, reflecting graded behavioural transitions rather than sharp class boundaries. Most misclassifications occurred at lower-severity boundaries (CN–MCI, MCI–MD), consistent with subtle and fluctuating early behavioural symptoms. The confusion matrix (Fig. ??) exhibited a structured, stage ordered pattern, with most misclassifications occurring between adjacent clinical stages rather than across non-contiguous severity levels.

Clinically, the most predictive features traced a coherent neuropsychiatric trajectory from early sleep disturbance and circadian dysregulation to escalating restlessness, caregiver distress, and hallucinations in advanced stages. Notably, this progression was recovered without cognitive, imaging, or biomarker inputs, yet aligned with established multimodal disease models. Within the framework, sleep behavioural markers provide a clinically meaningful, caregiver facing signal that complements molecular, structural, and linguistic indicators, particularly in real world settings where behavioural change often precedes measurable biological decline.

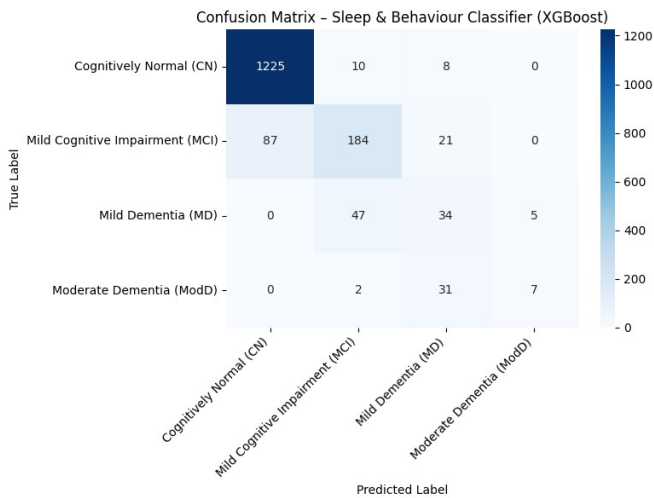


FIGURE 12 Confusion matrix for the sleep-behaviour classifier, illustrating prediction patterns across CN, MCI, MD, and MOD stages.

3.5 | Simulated Decision-Level Fusion: Cross-Modality Integration via Meta-Modeling

This component of the multimodal framework integrates heterogeneous Alzheimer's disease (AD) signals derived from molecular, structural, linguistic, and behavioural modalities into a unified stage prediction. Because the constituent datasets originate from independent ADNI sub-cohorts without subject-level alignment, multimodal integration was performed at the decision level rather than at the raw feature level. This design mirrors real-world clinical practice,

where laboratory assays, neuroimaging, cognitive assessments, and behavioural evaluations are often obtained asynchronously or incompletely, yet must be reconciled to form a coherent diagnostic judgment. This decision level fusion strategy constitutes the central methodological contribution of this study, enabling multimodal Alzheimer's disease severity modeling across independent cohorts without requiring subject-level alignment.

The fusion framework treats each unimodal classifier as an expert observer of a distinct biological or functional subsystem. Cerebrospinal-fluid biomarkers reflect underlying molecular processes associated with disease pathology, structural MRI captures patterns of neurodegeneration, language-derived features index executive and semantic network integrity, and sleep-behavioural measures quantify circadian disruption and neuropsychiatric instability. By learning how these expert opinions converge and diverge across disease stages, the meta-classifier produces an integrated severity estimate that preserves the biological interpretability of each modality while accommodating cross-modality heterogeneity.

3.5.1 | Fusion Data Construction and Supervision

A simulated fusion dataset was constructed using the predicted outputs from the four independently trained unimodal models described in Sections 3.1–3.4. For each simulated fusion instance, the combination of modality predictions—(i) the CSF biomarker model, (ii) the MRI model, (iii) the language-based model, and (iv) the sleep-behaviour model—was used to construct the fusion feature vector. Each element was encoded as an ordinal severity label reflecting the predicted clinical stage, ordered from lower to higher impairment: 0 = Cognitively Normal (CN), 1 = Mild Cognitive Impairment (MCI), 2 = Mild Dementia (MD), and 3 = Moderate Dementia (ModD).

Because subject level correspondence across modalities was unavailable, fusion samples were generated synthetically by aligning unimodal predictions to a common clinical reference. Structural MRI-derived Clinical Dementia Rating (CDR) stage labels were used as the supervisory ground truth for meta-classification, as MRI-based staging represents the most widely accepted clinical anchor for disease severity and reflects cumulative neurodegenerative burden. This supervision strategy ensured that all fused instances remained clinically interpretable and anchored to established diagnostic conventions while still allowing upstream and downstream modalities to influence decision boundaries.

3.5.2 | Exploratory Characterization of the Multimodal Decision Space

Prior to meta classification, the structure of the fused decision space was examined using the predicted stage outputs generated by the four unimodal models. Across the held-out test set ($n = 82$), the distribution

of predicted stage combinations revealed a clear stage-ordered structure. Early stage profiles exhibited greater variability across modalities, whereas mid stage cases showed stronger agreement among modality predictions. In contrast, advanced stage profiles demonstrated near complete convergence, with most modalities predicting the same or adjacent severity levels.

These patterns indicate that the fused decision space preserves a biologically coherent progression structure prior to meta classification, with variability concentrated in early disease stages and stronger consensus emerging as neurodegeneration advances.

3.5.3 | Preprocessing and Meta Classification Strategy

Only fusion instances containing predictions from all four modalities were retained to ensure consistent multimodal representation. After filtering incomplete cases, the final fusion dataset consisted of 410 multimodal prediction instances. Each fusion sample was represented by a four dimensional feature vector corresponding to the ordinal stage predictions produced by the CSF biomarker model, MRI model, language-based model, and sleep-behaviour model. Because these inputs correspond to discrete severity labels rather than continuous measurements, no normalization or feature scaling was required.

The dataset was divided using a stratified 80:20 train-test split to preserve the proportional distribution of disease stages across partitions while maintaining the ordinal severity structure inherent to the Alzheimer's disease continuum. This resulted in 328 training instances used for model learning and 82 held-out test instances reserved for evaluation.

A gradient-boosted XGBoost classifier was employed as the meta-classifier. The model was configured with 400 boosting trees, a maximum tree depth of 4, a learning rate of 0.05, subsampling of 0.9, column sampling of 0.9, and L2 regularization ($\lambda = 1.0$). These settings were selected to balance model capacity with generalization stability given the moderate size of the fusion dataset.

Decision level fusion involves non-linear interactions, modality-specific dominance patterns, and conditional dependencies that vary across disease stages. XGBoost efficiently captures such interactions and accommodates heterogeneous feature importance across severity levels while providing interpretable importance estimates reflecting the contribution of each modality. In contrast to linear meta models that assume additive relationships, gradient boosting enables the fusion model to learn stage-dependent decision boundaries reflecting the evolving contribution of molecular, structural, linguistic, and behavioural signals across the Alzheimer's disease continuum.

The trained meta classifier produces a unified disease-stage prediction together with associated class probabilities representing the integrated contribution of all modalities. This decision level fusion framework demonstrates how heterogeneous diagnostic signals can be reconciled at the decision level to generate a clinically interpretable

severity estimate even when subject level alignment across datasets is unavailable.

3.5.4 | Multimodal Decision-Level Fusion Results (Simulated Meta-Classifier)

The final experiment evaluated whether integrating modality-specific predictions, rather than raw features, could yield a more stable and biologically coherent clinical stage classifier for Alzheimer's disease progression. Decision-level fusion was implemented using probabilistic outputs from four independently trained models capturing cerebrospinal fluid biomarkers, structural MRI, language-derived features, and sleep-behavioural profiles. Because these modalities originated from non-overlapping cohorts, fusion was assessed in a simulated setting by aligning prediction distributions to construct plausible multimodal profiles. The XGBoost meta-classifier achieved strong and balanced performance (accuracy = 93.9%, macro-F1 = 0.94). Transitional stages were identified with high reliability, with perfect recall for both Mild Cognitive Impairment (MCI) and Moderate Dementia (ModD), indicating improved stability relative to unimodal models. Figure 13 presents the corresponding confusion matrix, which exhibits a predominantly diagonal structure, with misclassifications occurring primarily between adjacent stages, consistent with gradual and overlapping clinical progression. Mild Dementia (MD) demonstrated robust discrimination, while modest variability in Cognitively Normal (CN) cases was consistent with early behavioural and linguistic heterogeneity. Overall, decision-level integration reconstructed a clinically plausible severity continuum, demonstrating that heterogeneous diagnostic signals can be coherently combined to yield a more robust and clinically meaningful representation of Alzheimer's disease progression than any unimodal approach alone.

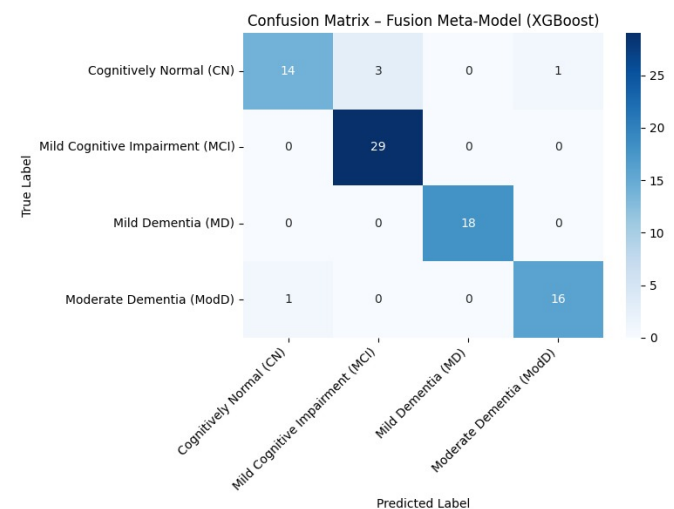


FIGURE 13 Confusion matrix for the sleep-behaviour classifier, illustrating prediction patterns across CN, MCI, MD, and ModD stages.

4 | CROSS-COHORT VALIDATION

To achieve clinical relevance, Alzheimer's disease models must preserve their predictive structure across populations that differ in linguistic context, demographics, acquisition protocols, and biomarker handling conditions. ADNI and CCNA represent two intentionally heterogeneous cohorts: ADNI follows U.S. clinical research standards with fixed MRI sequences, biomarker assay pipelines, and structured cognitive evaluations, whereas CCNA aggregates diverse Canadian regions with bilingual environments, varying scanner platforms, and distinct behavioural and sleep monitoring strategies. This diversity establishes a natural environment for evaluating whether a model captures core neurodegenerative signatures that transcend dataset-specific conditions. Because the cohorts share no participants, validation must assess the preservation of probability-space structure, rather than subject-level matches. The objective of this analysis is to determine whether independently trained ADNI and CCNA models express a cohort-invariant probabilistic representation of cognitive impairment—capturing consistent uncertainty structure, probability geometry, calibration behaviour, and severity ordering.

4.1 | External Generalization of the Multimodal Decision-Level Framework

To rigorously assess population-level generalizability, all unimodal and fusion models trained exclusively on ADNI were evaluated directly on the COMPASS-ND dataset. COMPASS-ND includes 1,426 clinically annotated visits, but with substantially fewer samples per modality, greater demographic heterogeneity, and non-uniform acquisition protocols. These characteristics make COMPASS-ND an appropriate test bed for evaluating whether the proposed *multimodal decision-level framework* captures fundamental disease-related structure rather than ADNI-specific patterns. Importantly, the exact same unimodal pipelines and fusion architecture used during ADNI training were applied to COMPASS-ND without any modification. This included identical preprocessing routines, feature engineering steps, classifier architectures (Random Forest for CSF, CNN for MRI, SVM for Language, and XGBoost for Sleep–Behaviour), and the decision-level fusion meta-classifier. No retraining, recalibration, hyperparameter adjustment, or domain adaptation was performed; therefore, COMPASS-ND performance reflects true external generalization.

External classification performance across modalities is summarized in Table 3. Sleep–behavioural signals achieved the highest external accuracy (71.4%), reflecting the robustness of circadian disturbance and perceptual-instability markers across populations and clinical settings. MRI and language modalities achieved identical accuracy (66.2%), demonstrating that structural atrophy patterns and quantitative linguistic impairment maintain moderate discriminative strength despite demographic, scanner, and linguistic variability. CSF biomarkers generalized less strongly (52.4%), likely influenced by assay-platform

TABLE 3 External Accuracy on COMPASS-ND

Modality	Accuracy (%)
MRI	66.2
Sleep	71.4
Language	66.2
CSF	52.4
Fusion	51.6

variability and differences in pre-analytical procedures. Fusion accuracy (51.6%) was lower, as COMPASS-ND lacks the balanced multimodal availability assumed by the simulated fusion model, highlighting the sensitivity of decision-level integration to missing-modality configurations in real-world clinical environments.

While accuracy provides an initial estimate of external performance, it does not capture whether deeper representational properties—such as uncertainty allocation, probability geometry, or disease-severity ordering—remain stable across cohorts. Accordingly, the following section evaluates cross-cohort alignment using a probability-space similarity framework.

4.2 | Probability-Space Similarity Framework

To evaluate whether models trained independently on ADNI and CCNA encode cohort-invariant internal reasoning rather than dataset-specific decision patterns, predicted probability vectors were compared using a suite of complementary statistical similarity measures. The objective was to determine whether two independently trained classifiers exhibit (i) a similar allocation of uncertainty across clinical stages, (ii) a similar probability-mass geometry, and (iii) preserved ordering along the continuum of severity of Alzheimer's disease (AD). Each metric isolates a distinct dimension of probabilistic behaviour; together, they provide a multidimensional assessment of cross-cohort stability.

(1) Jensen–Shannon Similarity (JSS): Jensen–Shannon similarity measures agreement in uncertainty allocation across diagnostic states and is robust to cohort-specific noise^{44,45}. JSS values lie in the range $[0, 1]$, where higher values indicate more similar uncertainty distributions and comparable interpretation of transitional impairment categories.

$$JSS = 1 - \sqrt{\frac{1}{2}KL(P \parallel M) + \frac{1}{2}KL(Q \parallel M)}, \quad M = \frac{1}{2}(P + Q)$$

(2) Bhattacharyya Coefficient (BC): The Bhattacharyya coefficient quantifies the overlap between two probability distributions by measuring the extent to which both models assign probability mass to the same impairment states⁴⁶. BC values lie in the range $[0, 1]$, with higher values indicating greater overlap between distributions.

$$BC(P, Q) = \sum_i \sqrt{P_i Q_i}$$

(3) Hellinger Similarity (HS): Hellinger similarity evaluates similarity in the shape of probability distributions, independent of absolute probability magnitude, and is sensitive to structural differences in uncertainty geometry⁴⁷. HS values lie in the range $[0, 1]$, where higher values indicate structurally similar uncertainty profiles across cohorts.

$$HS = 1 - \frac{1}{\sqrt{2}} \sqrt{\sum_i (\sqrt{P_i} - \sqrt{Q_i})^2}$$

(4) Pearson Similarity: Pearson similarity examines calibration coupling across cohorts by measuring the linear correlation between predicted probability vectors⁴⁸. Pearson correlation coefficients lie in the range $[-1, 1]$. Values close to $+1$ indicate strong positive calibration agreement, meaning that both models assign higher probabilities to the same disease stages. Values near 0 indicate little or no linear relationship between probability assignments across cohorts. Values approaching -1 indicate inverse calibration behaviour, where stages assigned higher probability by one model tend to receive lower probability from the other.

$$\rho = \frac{\text{cov}(P, Q)}{\sigma_P \sigma_Q}$$

(5) Kendall's τ : Kendall's τ assesses preservation of ordinal severity ordering by providing a non-parametric measure of whether two models rank individuals consistently along the CN $\rightarrow$ MCI $\rightarrow$ Mild $\rightarrow$ Moderate Dementia continuum⁴⁹. Kendall's τ lies in the range $[-1, 1]$. Values close to $+1$ indicate strong agreement in severity ordering across cohorts, values near 0 indicate no consistent ordinal relationship, and values approaching -1 indicate reversed ordering, where individuals ranked as more severe by one model tend to be ranked as less severe by the other.

$$\tau = \frac{C - D}{\frac{1}{2}n(n-1)} \quad (1)$$

(6) Composite Probability-Space Similarity (R_{prob}): To summarize overall distributional agreement between cohorts, a composite probability-space similarity index (R_{prob}) was computed as the mean of Jensen-Shannon similarity (JSS), the Bhattacharyya coefficient (BC), Pearson similarity (ρ), and Hellinger similarity (HS). These metrics capture complementary aspects of probability alignment, including uncertainty allocation, probability-mass overlap, calibration coupling, and distributional geometry.

Higher values of R_{prob} indicate stronger agreement in the probabilistic representation of disease stages across cohorts, whereas lower values indicate weaker similarity or divergence in probability-space structure.

$$R_{\text{prob}} = \frac{JSS + BC + \rho + HS}{4} \quad (2)$$

Composite Alignment Index (R_{all}). To jointly capture probabilistic similarity and ordinal severity preservation, a composite alignment index (R_{all}) was defined by integrating R_{prob} with Kendall's τ . R_{all} lies in the range $[-1, 1]$, where higher values indicate stronger global alignment across cohorts.

$$R_{\text{all}} = \frac{R_{\text{prob}} + \tau}{2} \quad (3)$$

Robustness of the similarity estimates was assessed using bootstrap resampling with 1,000 iterations. In each iteration, probability vectors were sampled with replacement and all similarity metrics were recomputed. For each modality, the bootstrap mean of R_{prob} provides a stable estimate of the expected similarity value across resampled datasets. The corresponding 95% confidence interval (CI) represents the empirical range containing 95% of bootstrap estimates and quantifies the uncertainty associated with the similarity measure. Narrow confidence intervals indicate stable cross-cohort similarity estimates, whereas wider intervals suggest greater variability due to sampling differences.

4.3 | Modality Wise Cross Cohort Similarity

To isolate how each modality behaves under cross-cohort variation, the probability-space similarity framework was applied to MRI, Sleep, Language, CSF, and Fusion models individually. This provides a multidimensional view of which modalities encode cohort-invariant structure and which are influenced by technical or demographic differences. Quantitative cross-cohort similarity metrics for each modality are summarized in Table 4.

TABLE 4 Cross-cohort probability-space similarity metrics (ADNI $\leftrightarrow$ CCNA).

Metric	Lang	Slp	CSF	MRI	Fus
Jensen-Shannon Sim. (JSS)	0.677	0.794	0.692	0.918	0.701
Bhattacharyya Coef. (BC)	0.689	0.831	0.725	0.904	0.900
Pearson (ρ)	0.559	0.502	0.532	0.832	0.781
Hellinger Sim. (HS)	0.667	0.677	0.583	0.690	0.683
Kendall's τ	-0.131	0.021	0.068	0.333	0.333
Composite R_{prob}	0.648	0.701	0.633	0.836	0.766
Composite R_{all}	0.492	0.565	0.520	0.585	0.680
Bootstrap R_{prob} mean	0.648	0.701	0.635	0.819	0.766
Bootstrap 95% CI	0.64-0.66	0.68-0.72	0.55-0.72	0.70-0.89	0.65-0.78

4.4 | Integrated Interpretation of Cross-Cohort Stability

The cross-cohort similarity metrics reveal a consistent and biologically grounded pattern across modalities. Structural MRI shows the strongest alignment, with the highest composite similarity ($R_{\text{prob}} = 0.836$) and excellent agreement across all probability-space dimensions (JSS = 0.918, BC = 0.904, Pearson = 0.832). These results indicate that neurodegenerative signatures—particularly cortical atrophy and ventricular expansion—generalize well across populations despite differences in scanner platforms, sampling frameworks, and demographic distributions.

Fusion models also achieve high cross-cohort coherence ($R_{\text{prob}} = 0.766$), closely approaching MRI. This suggests that combining structural,

behavioural, linguistic, and biochemical cues yields a consolidated representation that remains stable even when reconstructed in the independent CCNA dataset. The strong overlap in probability geometry ($BC = 0.900$) further confirms that multimodal integration reinforces shared diagnostic structure rather than amplifying cohort-specific noise.

Sleep-behaviour signals demonstrate reliable generalizability, with substantial preservation of uncertainty allocation and probability-mass structure ($R_{\text{prob}} = 0.701$). The high Bhattacharyya coefficient ($BC = 0.831$) indicates that circadian irregularities and nighttime disruption, captured through Neuropsychiatric Inventory (NPI) trajectories, encode physiologically conserved changes that remain interpretable across cohorts with different environmental and lifestyle profiles.

Language-based models maintain moderate alignment ($R_{\text{prob}} = 0.648$), supported by robust JSS (0.677) and BC (0.689) values. However, ordinal ranking consistency is lower ($\tau = -0.131$), which is consistent with expected variation in linguistic context, bilingualism, and educational exposure across ADNI and CCNA populations. Although the underlying distributional structure is preserved, rank-order behaviour likely reflects cohort-level differences in speech production and language use.

CSF biomarkers show moderate cross-cohort coherence ($R_{\text{prob}} = 0.633$). Core biochemical relationships such as amyloid-tau coupling remain evident in the overlap metrics ($BC = 0.725$), but assay-specific variability and differences in pre-analytical handling reduce calibration similarity (Pearson = 0.532). Consequently, CSF biomarkers retain biologically meaningful signal but exhibit lower cross-cohort stability compared with imaging and behavioural indicators.

Overall, the integrated pattern demonstrates that the proposed multimodal decision-level framework captures cohort-resilient and biologically meaningful probability structures rather than dataset-specific patterns. Across uncertainty allocation, probability geometry, calibration behaviour, and severity ordering, models trained on ADNI and evaluated on CCNA exhibit consistent internal reasoning despite differences in acquisition design, demographic characteristics, and biomarker pipelines. Structural MRI and multimodal fusion provide the most stable backbone for cross-cohort generalization, while sleep-behavioural and language modalities contribute reliable non-invasive signals. Bootstrap analyses further confirm that these similarity patterns remain stable across resampled datasets.

5 | DISCUSSION

This study investigated whether heterogeneous Alzheimer's disease markers—spanning molecular, structural, linguistic, and behavioural domains—can be reconciled into a coherent severity representation despite originating from distinct cohorts and measurement frameworks. The results demonstrate that each modality encodes a biologically meaningful but incomplete view of disease progression, and that their integration at the decision level yields a more stable and interpretable staging geometry than any unimodal model alone.

Answer to RQ-1: Alzheimer's disease severity can be characterized through the integration of complementary biological and behavioural signals spanning molecular, structural, cognitive, and behavioural domains. The experimental results demonstrate that CSF biomarkers capture early biochemical pathology, structural MRI reflects progressive neurodegeneration, language markers reveal early cognitive impairment, and sleep-behavioural disturbances reflect downstream functional disruption. Together, these modalities provide a multidimensional characterization of disease severity across the CN-MCI-MD-ModD continuum.

Across unimodal experiments, modality-specific strengths aligned with established Alzheimer's disease pathophysiology. Structural MRI provided the most stable separation in mid-to-late disease stages, reflecting cumulative and irreversible neurodegeneration. CSF biomarkers captured upstream amyloid-tau dysregulation and were particularly sensitive at early biochemical transition points preceding overt structural decline. Language markers proved most informative for early impairment, consistent with vulnerability of frontal-temporal cognitive networks, while sleep and behavioural disturbances reflected downstream circadian dysregulation and neuropsychiatric instability that often emerge during disease progression.

Answer to RQ-2: Different indicators of Alzheimer's disease exhibit stage-dependent patterns of progression aligned with established disease models. Structural MRI provides the most stable separation in mid-to-late disease stages, CSF biomarkers capture early amyloid-tau dysregulation, language markers reveal early cognitive decline, and sleep-behavioural disturbances reflect downstream circadian and neuropsychiatric disruption. These findings confirm that different clinical indicators reflect distinct yet ordered components of disease progression.

These complementary gradients directly address the central question posed in the Introduction regarding whether different modalities capture distinct aspects of Alzheimer's disease severity. Together, the results demonstrate that molecular, structural, cognitive-linguistic, and behavioural indicators encode partially overlapping yet systematically ordered components of the neurodegenerative process. A key finding of this study is that severity information remains preserved in probability space even when modalities originate from non-overlapping cohorts and independent measurement pipelines. The simulated fusion experiment demonstrates that decision-level integration can reconstruct a clinically plausible progression trajectory without requiring subject-level multimodal alignment. This finding is particularly important in real-world clinical settings, where complete multimodal records are rarely available for every patient. By integrating probabilistic outputs rather than raw features, the proposed framework preserves the diagnostic semantics of each modality while enabling coherent cross-modality reasoning. The resulting severity representation therefore reflects convergent impairment signals across biological

and behavioural systems rather than modality-specific artefacts, supporting the use of probability-space modeling as a robust strategy for multimodal Alzheimer's disease staging.

Answer to RQ-3: Different sources of clinical evidence contribute complementary perspectives on Alzheimer's disease severity. While unimodal models capture meaningful but partial representations of impairment, the decision-level fusion model integrates heterogeneous diagnostic signals to produce a more stable and interpretable severity representation. The fusion framework therefore provides a unified representation of disease progression by reconciling molecular, structural, linguistic, and behavioural evidence.

Importantly, the fusion model did not simply average unimodal predictions but instead learned structured interactions among modalities. Early-stage variability was driven primarily by language and behavioural signals, mid-stage ambiguity reflected asynchronous molecular and structural progression, and late-stage convergence was dominated by agreement between MRI and CSF signals. This staged interaction pattern mirrors established biological models of Alzheimer's disease progression and provides mechanistic interpretability beyond raw classification accuracy.

Answer to RQ-4: Cross-cohort validation demonstrates that the proposed multimodal framework preserves a consistent probabilistic representation of Alzheimer's disease severity across independent datasets. Probability-space similarity analysis between ADNI and CCNA revealed strong alignment ($R_{\text{prob}} \approx 0.77$), indicating that the learned staging structure reflects underlying neurodegenerative mechanisms rather than dataset-specific artefacts.

This cross-cohort consistency suggests that the proposed framework captures cohort-resilient patterns of disease progression rather than relying on dataset-specific characteristics. Together, these findings support a multimodal severity representation in which molecular, neurodegenerative, linguistic, and behavioural signals jointly characterize Alzheimer's disease progression. Rather than proposing a new diagnostic taxonomy, the framework demonstrates how heterogeneous markers can be organized into a unified, biologically coherent severity space.

6 | LIMITATIONS

The present study was conducted under a deliberately controlled experimental design to ensure strong internal validity and clear interpretability of modality-specific signals. Unimodal models were trained on harmonized research datasets with standardized preprocessing and well-defined stage labels, allowing each modality's impairment geometry to be characterized with minimal confounding.

While this structure supports reliable methodological comparison, it may yield optimistic performance estimates relative to real-world clinical environments, particularly for behavioural and language modalities where symptom expression and reporting can be variable. These effects were partially mitigated through stratified evaluation, conservative oversampling, and emphasis on stage geometry rather than raw accuracy. With respect to external and construct validity, the decision-level fusion framework intentionally operates without subject-level multimodal overlap, reflecting realistic clinical scenarios in which complete data are rarely available. As a result, fusion was implemented using aligned probability outputs rather than raw features, enabling principled integration while preserving biological semantics across cohorts. Similarly, behavioural and language stages were derived using clinically grounded proxies rather than diagnostic labels, prioritizing interpretability and longitudinal coherence. These design choices position the framework as a scalable and flexible methodological foundation, while naturally motivating future validation on fully multimodal, patient-level longitudinal datasets.

7 | CONCLUSION

In summary, this study presents a multidimensional framework for representing Alzheimer's disease severity by integrating structural imaging, cerebrospinal fluid biomarkers, language-derived measures, and sleep-behavioural indicators. Across heterogeneous cohorts and modalities, the framework demonstrates stable severity alignment, indicating that it captures underlying neurodegenerative patterns rather than dataset-specific artefacts. Structural MRI provides the most robust staging signals, while language, sleep-behavioural measures, and CSF biomarkers contribute complementary sensitivity at different phases of impairment. Future extensions of this work may incorporate longitudinal trajectories, blood-based biomarkers, and digital behavioural signals to enhance early detection and real-world applicability. Prospective validation in clinical settings will be essential to assess translational feasibility and clinical impact.

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